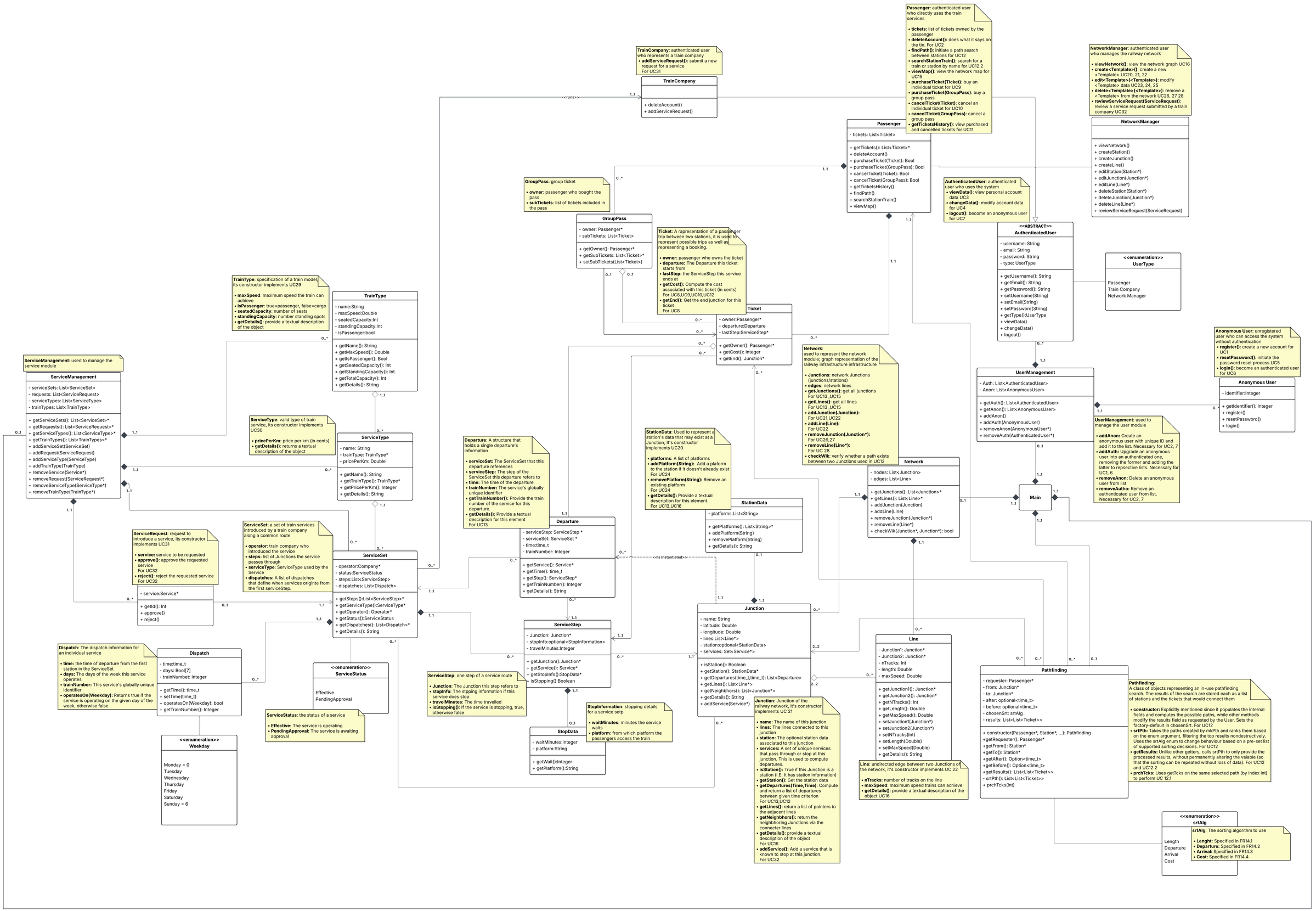
TrainCompany: authenticated user who represents a train company
• addServiceRequest(): submit a new request for a service
For UC31
TrainCompany
+ deleteAccount()
+ addServiceRequest()
<<runs>>
1..1
Passenger: authenticated user who directly uses the train services
• tickets: list of tickets owned by the passenger
• deleteAccount(): does what it says on the tin. For UC2
• findPath(): initiate a path search between stations for UC12
• searchStationTrain(): search for a train or station by name for UC12.2
• viewMap(): view the network map for UC15
• purchaseTicket(Ticket): buy an individual ticket for UC9
• purchaseTicket(GroupPass): buy a group pass
• cancelTicket(Ticket): cancel an individual ticket for UC10
• cancelTicket(GroupPass): cancel a group pass
• getTicketsHistory(): view purchased and cancelled tickets for UC11
NetworkManager: authenticated user who manages the railway network
• viewNetwork(): view the network graph UC16
• create<Template>(): create a new <Template> UC20, 21, 22
• edit<Template>(<Template>): modify <Template> data UC23, 24, 25
• delete<Template>(<Template>): remove a <Template> from the network UC26, 27 28
• reviewServiceRequest(ServiceRequest): review a service request submitted by a train company UC32
Passenger
- tickets: List<Ticket>
+ getTickets(): List<Ticket>*
+ deleteAccount()
+ purchaseTicket(Ticket): Bool
+ purchaseTicket(GroupPass): Bool
+ cancelTicket(Ticket): Bool
+ cancelTicket(GroupPass): Bool
+ getTicketsHistory()
+ findPath()
+ searchStationTrain()
+ viewMap()
NetworkManager
+ viewNetwork()
+ createStation()
+ createJunction()
+ createLine()
+ editStation(Station*)
+ editJunction(Junction*)
+ editLine(Line*)
+ deleteStation(Station*)
+ deleteJunction(Junction*)
+ deleteLine(Line*)
+ reviewServiceRequest(ServiceRequest)
GroupPass: group ticket
• owner: passenger who bought the pass
• subTickets: list of tickets included in the pass
0..*
1..1
AuthenticatedUser: authenticated user who uses the system
• viewData(): view personal account data UC3
• changeData(): modify account data for UC4
• logout(): become an anonymous user for UC7
GroupPass
- owner: Passenger*
- subTickets: List<Ticket>
+ getOwner(): Passenger*
+ getSubTickets: List<Ticket>*
+ setSubTickets(List<Ticket>)
Ticket: A rapresentation of a passenger trip between two stations, it is used to represent possible trips as well as representing a booking.
• owner: passenger who owns the ticket
• departure: The Departure this ticket starts from
• lastStep: the ServiceStep this service ends at
• getCost(): Compute the cost associated with this ticket (in cents)
For UC8,UC9,UC10,UC12
• getEnd(): Get the end junction for this ticket
For UC8
<<ABSTRACT>>
AuthenticatedUser
- username: String
- email: String
- password: String
- type: UserType
+ getUsername(): String
+ getEmail(): String
+ getPassword(): String
+ setUsername(String)
+ setEmail(String)
+ setPassword(String)
+ getType():UserType
+ viewData()
+ changeData()
+ logout()
<<enumeration>>
UserType
Passenger
Train Company
Network Manager
0..1
0..1
Ticket
- owner:Passenger*
- departure:Departure
- lastStep:ServiceStep*
+ getOwner(): Passenger*
+ getCost(): Integer
+ getEnd(): Junction*
0..*
0..*
0..*
0..*
0..*
0..*
0..*
1..1
TrainType: specification of a train model, its constructor implements UC29
• maxSpeed: maximum speed the train can achieve
• isPassenger: true=passenger, false=cargo
• seatedCapacity: number of seats
• standingCapacity: number standing spots
• getDetails(): provide a textual description of the object
TrainType
- name:String
- maxSpeed:Double
- seatedCapacity:Int
- standingCapacity:Int
- isPassenger:bool
+ getName(): String
+ getMaxSpeed(): Double
+ getIsPassenger(): Bool
+ getSeatedCapacity(): Int
+ getStandingCapacity(): Int
+ getTotalCapacity(): Int
+ getDetails(): String
Network:
used to represent the network module; graph representation of the railway infrastructure infrastructure
• Junctions: network Junctions (junctions/stations)
• edges: network lines
• getJunctions(): get all junctions
For UC13 ,UC15
• getLines(): get all lines
For UC13 ,UC15
• addJunction(Junction):
For UC21,UC22
• addLine(Line):
For UC22
• removeJunction(Junction*):
For UC26,27
• removeLine(Line*):
For UC 28
• checkWlk: verify whether a path exists between two Junctions used in UC12
Anonymous User: unregistered user who can access the system without authentication
• register(): create a new account for UC1
• resetPassword(): initiate the password reset process UC5
• login(): become an authenticated user for UC6
Anonymous User
- identifier:Integer
+ getIdentifier(): Integer
+ register()
+ resetPassword()
+ login()
UserManagement
- Auth: List<AuthenticatedUser>
- Anon: List<AnonymousUser>
+ getAuth(): List<AuthenticatedUser>
+ getAnon(): List<AnonymousUser>
+ addAnon()
+ addAuth(AnonymousUser)
+ removeAnon(AnonymousUser*)
+ removeAuth(AuthenticatedUser*)
1..1
0..*
0..1
1..1
UserManagement: used to manage the user module
• addAnon: Create an anonymous user with unique ID and add it to the list. Necessary for UC2, 7
• addAuth: Upgrade an anonymous user into an authenticated one, removing the former and adding the latter to repsective lists. Necessary for UC1, 6
• removeAnon: Delete an anonymous user from list
• removeAutho: Remove an authenticated user from list.
Necessary for UC2, 7
ServiceManagement: used to manage the service module
ServiceManagement
- serviceSets: List<ServiceSet>
- requests: List<ServiceRequest>
- serviceTypes: List<ServiceType>
- trainTypes: List<TrainType>
+ getServiceSets(): List<ServiceSet>*
+ getRequests(): List<ServiceRequest>*
+ getServiceTypes(): List<ServiceType>*
+ getTrainTypes(): List<TrainTypes>*
+ addServiceSet(ServiceSet)
+ addRequest(ServiceRequest)
+ addServiceType(ServiceType)
+ addTrainType(TrainType)
+ removeService(Service*)
+ removeRequest(ServiceRequest*)
+ removeServiceType(ServiceType*)
+ removeTrainType(TrainType*)
0..1
1..1
1..1
1..1
1..1
0..*
ServiceType: valid type of train service, its constructor implements UC30
• pricePerKm: price per km (in cents)
• getDetails(): returns a textual description of the object
ServiceType
- name: String
- trainType: TrainType*
- pricePerKm: Double
+ getName(): String
+ getTrainType(): TrainType*
+ getPricePerKm(): Integer
+ getDetails(): String
1..1
0..*
Main
1..1
1..1
1..1
1..1
Departure: A structure that holds a single departure's information
• serviceSet: The ServiceSet that this departure references
• serviceStep: The step of the ServiceSet this departure refers to
• time: The time of the departure
• trainNumber: The service's globally unique identifier
• getTrainNumber(): Provide the train number of the service for this departure.
• getDetails(): Provide a textual description for this element
For UC13
StationData: Used to represent a station's data that may exist at a Junction, It's constructor implements UC20
• platforms: A list of platforms
• addPlatform(String): Add a plaform to the station if it doesn't already exist
For UC24
• removePlatform(String): Remove an existing platform
For UC24
• getDetails(): Provide a textual description for this element.
For UC13,UC16
Network
- nodes: List<Junction>
- edges: List<Line>
+ getJunctions(): List<Junction>*
+ getLines(): List<Line>*
+ addJunction(Junction)
+ addLine(Line)
+ removeJunction(Junction*)
+ removeLine(Line*)
+ checkWlk(Junction*, Junction*): bool
StationData
- platforms:List<String>
+ getPlatforms(): List<String>*
+ addPlatform(String)
+ removePlatform(String)
+ getDetails(): String
1..1
0..1
1..1
1..1
1..1
1..1
ServiceRequest: request to introduce a service, its constructor implements UC31
• service: service to be requested
• approve(): approve the requested service
For UC32
• reject(): reject the requested service
For UC32
ServiceSet: a set of train services introduced by a train company along a common route
• operator: train company who introduced the service
• steps: list of Junctions the service passes through
• serviceType: ServiceType used by the Service
• dispatches: A list of dispatches that define when services originte from the first serviceStep.
Departure
- serviceStep: ServiceStep *
- serviceSet: ServiceSet *
- time:time_t
- trainNumber: Integer
+ getService(): Service*
+ getTime(): time_t
+ getStep(): ServiceStep*
+ getTrainNumber(): Integer
+ getDetails(): String
<<Is Instantiated>>
0..*
0..*
0..*
0..*
0..*
0..*
1..1
1..1
ServiceSet
- service:Service*
+ getId(): Int
+ approve()
+ reject()
ServiceSet
- operator:Company*
- status:ServiceStatus
- steps:List<ServiceStep>
- dispatches: List<Dispatch>
+ getSteps():List<ServiceStep>*
+ getServiceType():ServiceType*
+ getOperator(): Operator*
+ getStatus():ServiceStatus
+ getDispatches(): List<Dispatch>*
+ getDetails(): String
0..*
0..1
1..1
1..1
1..1
1..1
0..*
ServiceStep
- Junction: Junction*
- stopInfo:optional<StopInformation>
- travelMinutes:Integer
+ getJunction():Junction*
+ getService(): Service*
+ getStopInfo():StopData*
+ isStopping():Boolean
1..1
1..1
0..*
0..*
1..1
Junction
- name: String
- latitude: Double
- longitude: Double
- lines:List<Line*>
- station:optional<StationData>
- services: Set<Service*>
+ isStation(): Boolean
+ getStation(): StationData*
+ getDepartures(time_t,time_t): List<Departure>
+ getLines(): List<Line*>
+ getNeighbhors(): List<Junction*>
+ getDetails(): String
+ addService(Service*)
0..*
2..2
2..2
0..*
Line
- Junction1: Junction*
- Junction2: Junction*
- nTracks: Int
- length: Double
- maxSpeed: Double
+ getJunction1(): Junction*
+ getJunction2(): Junction*
+ getNTracks(): Int
+ getLength(): Double
+ getMaxSpeed(): Double
+ setJunction1(Junction*)
+ setJunction2(Junction*)
+ setNTracks(Int)
+ setLength(Double)
+ setMaxSpeed(Double)
+ getDetails(): String
0..*
0..*
0..*
0..*
0..1
Dispatch: The dispatch information for an individual service
• time: the time of departure from the first station in the ServiceSet
• days: The days of the week this service operates
• trainNumber: This service's globally unique identifier
• operatesOn(Weekday): Returns true if the service is operating on the given day of the week, otherwise false
Dispatch
- time:time_t
- days: Bool[7]
- trainNumbet: Integer
+ getTime(): time_t
+ setTime(time_t)
+ operatesOn(Weekday): bool
+ getTrainNumber(): Integer
0..*
<<enumeration>>
ServiceStatus
Effective
PendingApproval
ServiceStatus: the status of a service
• Effective: The service is operating
• PendingApproval: The service is awaiting approval
ServiceStep: one step of a service route
• Junction: The Junction this step refers to
• stopInfo: The stpping information if this service does stop
• travelMinutes: The time travelled
• isStopping(): If the service is stopping, true, otherwise false
StopInformation: stopping details for a service setp
• waitMinutes: minutes the service waits
• platform: from which platform the passengers access the train
<<enumeration>>
Weekday
Monday = 0
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday = 6
StopData
- waitMinutes:Integer
- platform:String
+ getWait():Integer
+ getPlatform():String
0..1
Junction: Junction of the railway network, it's constructor implements UC 21
• name: The name of this junction
• lines: The lines connected to this junction
• station: The optional station data associated to this junction
• services: A set of unique services that pass through or stop at this junction. This is used to compute departures.
• isStation(): True if this Junction is a station (I.E. It has station information)
• getStation(): Get the station data
• getDepartures(Time,Time): Compute and return a list of departures between given time criterion
For UC13,UC12
• getLines(): return a list of pointers to the adjacent lines
• getNeighbhors(): return the neighbhoring Junctions via the connecter lines
• getDetails(): provide a textual description of the object
For UC16
• addService(): Add a service that is known to stop at this junction.
For UC32
Line: undirected edge between two Junctions of the network, it's constructor implements UC 22
• nTracks: number of tracks on the line
• maxSpeed: maximum speed trains can achieve
• getDetails(): provide a textual description of the object UC16
Pathfinding
- requester: Passenger*
- from: Junction*
- to: Junction*
- after: optional<time_t>
- before: optional<time_t>
- chosenSrt: srtAlg
- results: List<List<Ticket>>
+ constructor(Passenger*, Station*, ...): Pathfinding
+getRequester(): Passenger*
+getFrom(): Station*
+getTo(): Station*
+getAfter(): Option<time_t>
+getBefore(): Option<time_t>
+getResults(): List<List<Ticket>>
- srtPth(): List<List<Ticket>>
+ prchTcks(int)
Pathfinding:
A class of objects representing an in-use pathfinding search. The results of the search are stored each as a list of stations and the tickets that would connect them
• constructor: Explicitly mentioned since it populates the internal fields and computes the possible paths, while other methods modify the results field as requested by the User. Sets the factory-default in chosenSrt. For UC12
• srtPth: Takes the paths created by mkPth and ranks them based on the enum argument, filtering the top results nondestructively. Uses the srtAlg enum to change behaviour based on a pre-set list of supported sorting decisions. For UC12
• getResults: Unlike other getters, calls srtPth to only provide the processed results, without permanently altering the valable (so that the sorting can be repeated without loss of data). For UC12 and UC12.2
• prchTcks: Uses getTcks on the same selected path (by index int) to perform UC 12.1
<<enumeration>>
srtAlg
Length
Departure
Arrival
Cost
srtAlg: The sorting algorithm to use
• Lenght: Specified in FR14.1
• Departure: Specified in FR14.2
• Arrival: Specified in FR14.3
• Cost: Specified in FR14.4